

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} \quad (M\vec{u} = \vec{f})$$

$$\Leftrightarrow \begin{cases} Ax + By = a & \dots \textcircled{1} \\ Cx + Dy = b & \dots \textcircled{2} \end{cases}$$

$$BD^{-1} \times \textcircled{2} : (\text{Assume } D \text{ invertible})$$

↳ this is also the condition for the existence of Schur complement

$$BD^{-1}Cx + By = BD^{-1}b \dots \textcircled{3}$$

$$\textcircled{1} - \textcircled{3} : \underline{(A - BD^{-1}C)}x = a - BD^{-1}b$$

↳ M/D , Schur complement of block D

$$x = (M/D)^{-1}(a - BD^{-1}b)$$

$$y = D^{-1}(b - Cx) = D^{-1}(b - C(M/D)^{-1}(a - BD^{-1}b))$$

∴ solve original $M\vec{u} = \vec{f}$, NT invert M (dim-n)

Now we NT invert D and M/D (dim- $\frac{n}{2}$)